

PATENT RESEARCH REPORT

1. ABSTRACT

This technical report synthesizes patent evidence for Low Acrylonitrile Nitrile Butadiene Rubber (NBR), analyzing six primary patents: US9650502, US9034992, US7923518, US8389623, US8436091, and US8664340. The research scope encompasses emulsion polymerization, latex coagulation, and subsequent compounding/vulcanization processes to optimize the physical and mechanical properties of nitrile rubbers. Key technical approaches include cold emulsion polymerization (typically at 7-8 °C) using redox initiation systems (such as p-menthane hydroperoxide with iron(II) salts) and molecular weight regulators (tertiary dodecyl mercaptan, TDM) added incrementally to control conversion and polymer structure. A major process trend identified is the precise control of latex coagulation conditions, specifically the choice of coagulant salt (magnesium chloride, calcium chloride, or sodium chloride) and pH adjustment (using potassium hydroxide or hydrochloric acid). These coagulation parameters dictate the residual ion indices in the solid rubber, which directly influence Mooney scorch, vulcanization rates (t90 - t10), and vulcanizate mechanical properties such as stress at 30% and 300% elongation. The analyzed patents demonstrate that optimizing the cation ratios and emulsifier washing steps yields rubbers with superior aging resistance, lower mold corrosion, and enhanced tensile strength, providing a comprehensive framework for the synthesis and processing of high-performance low-acrylonitrile NBR.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

Patent Details - Patent Number: US9650502 - Patent Title: Process for the preparation of nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2017 - Material Type: Nitrile Rubber Latex - Relevance to target: Not disclosed

Polymerization and Coagulation Method Modifier (TDM) was added incrementally (at 37% and 59% conversion) to regulate molecular weight. Polymerization was stopped at 70% conversion using hydroxylamine sulfate. Latex was coagulated using magnesium sulfate, washed at pH 11.5-11.8 (adjusted with KOH), and dried at 100 °C.

Relevant Experimental Evidence - Latex A was prepared with 500 g acrylonitrile and 4.37 kg butadiene, resulting in a bound ACN content of 28.1% and Mooney viscosity of 60.5. - Latex B was prepared with 815 g acrylonitrile and 4.0 kg butadiene, resulting in a bound ACN content of

33.9% and Mooney viscosity of 32.1. - Latex C resulted in a bound ACN content of 27.9% and Mooney viscosity of 29.0. - Coagulation of 10 kg latex was achieved using 43 kg water and 4 kg saturated magnesium sulfate solution, followed by alkaline washing and drying.

Technical Relevance Provides a detailed recipe for synthesizing low-ACN nitrile rubber latexes (27.9% to 33.9% bound ACN) with controlled Mooney viscosity using incremental modifier additions.

Patent 2

Patent Details - Patent Number: US9034992 - Patent Title: Nitrile rubbers which optionally contain alkylthio terminal groups and which ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2015 - Material Type: Compounded Nitrile Rubber - Relevance to target: Not disclosed

Compounding and Vulcanization Method

Relevant Experimental Evidence - Compounding was performed in a 1.5 L laboratory internal mixer using nitrile rubbers from Example series 4. - Vulcanization was conducted at 160 °C under 120 bar of hydraulic pressure for 30 minutes. - Latex coagulation was performed using a magnesium chloride solution containing acid-ashed gelatin (10% strength, viscosity 92.1 cP, IEP 8.7). - Corrosion testing of sealing rings was conducted under 80% relative atmospheric humidity at 70 °C between sand-blasted aluminum plates.

Technical Relevance Focuses on the post-polymerization compounding, vulcanization, and performance testing (aging, corrosion, tensile properties) of nitrile rubbers.

Patent 3

Patent Details - Patent Number: US7923518 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2011 - Material Type: Nitrile Rubber - Relevance to target: Not disclosed

Technical Method

Relevant Experimental Evidence - The patent provides foundational disclosures regarding nitrile rubber compositions and their general properties. - No specific experimental parameters were extracted in the source data, but it belongs to the same family of nitrile rubber technology focusing on polymer properties.

Technical Relevance Provides foundational context for the composition and properties of nitrile rubbers within the patent family.

Patent 4

Patent Details - Patent Number: US8389623 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2013 - Material Type: Compounded Nitrile Rubber - Relevance to target: Not disclosed

Compounding and Coagulation Method

Relevant Experimental Evidence - Nitrile rubbers were processed in a 1.5 L mechanical laboratory mixer with constituents reported in parts by weight per 100 parts of rubber. - Vulcanization was performed at 160 °C under 120 bar of hydraulic pressure for 30 minutes. - NBR latices coagulated with magnesium chloride displayed lower Mooney scorch, shorter vulcanization times, and higher stress at 300% elongation compared to non-invention rubbers.

Technical Relevance Demonstrates how the choice of coagulant (magnesium chloride) and resulting cation ratios in the rubber improve Mooney scorch and vulcanization rates.

Patent 5

Patent Details - Patent Number: US8436091 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2013 - Material Type: Nitrile Rubber - Relevance to target: Not disclosed

Polymerization Method

Relevant Experimental Evidence - Nitrile rubbers G and H were synthesized using 61.4 parts butadiene, 36 parts acrylonitrile, and 2.6 parts 2-hydroxyethyl methacrylate at 8.0 °C. - The use of a specific redox system containing FeCl₂ (Rubber H) instead of FeSO₄ (Rubber G) resulted in a polymerization conversion of 84.2% in 6.5 hours compared to 86.1% in 7.0 hours. - Solid rubbers obtained exhibited Mooney viscosities ranging from 25 to 142 MU and glass transition temperatures (T_g) between -25.8 °C and -23.1 °C.

Technical Relevance Discloses emulsion polymerization of nitrile rubbers using a specific redox system (Fe(II) chloride vs Fe(II) sulfate) to control polymer properties like Mooney viscosity and glass transition temperature.

Patent 6

Patent Details - Patent Number: US8664340 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2014 - Material Type: Nitrile Rubber - Relevance to target: Not disclosed

Coagulation and Compounding Method

Relevant Experimental Evidence - Latex pH was adjusted using 5% strength aqueous KOH or HCl prior to coagulation. - Coagulation was performed using either NaCl (15-26 wt%) or CaCl₂ (0.3-1.2 wt%) at temperatures of 20-50 °C, followed by washing for 2.5-9.0 hours. - Nitrile rubbers coagulated with CaCl₂ exhibited higher ion indices (88.5-96.2) and longer t₉₀ times compared to

those coagulated with NaCl (16.9-44.1). - Vulcanizates with controlled ion indices showed a higher vulcanization rate (t90 - t10) and higher stress values at 30% elongation (up to 10.2 MPa for CaCl2-coagulated rubbers).

Technical Relevance Focuses on the coagulation of nitrile rubber latexes using specific salts (NaCl vs CaCl2) and pH adjustments to control ion indices, which directly affect vulcanization rates and vulcanizate properties.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

Comparison Table: | Patent | Polymerization Temperature | Target Conversion | Laboratory Mixer Volume | Bound ACN Content | Key Finding | |---|---|---|---|---|---| | US9650502 | Not disclosed | Not disclosed | Not disclosed | Not disclosed | | | US9034992 | Not disclosed | Not disclosed | Not disclosed | Not disclosed | | | US7923518 | Not disclosed | Not disclosed | Not disclosed | Not disclosed | | | US8389623 | Not disclosed | Not disclosed | Not disclosed | Not disclosed | | | US8436091 | Not disclosed | Not disclosed | Not disclosed | Not disclosed | | | US8664340 | Not disclosed | Not disclosed | Not disclosed | Not disclosed | | - No cross-patent comparison provided.

4. REFERENCES

- No references provided.